Partitioning $\{1, \ldots, n\}$ into parts of size at most three whose sums are powers of three

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Abstract

We prove the conjecture of Gurvich and Naumova that for every $n \geq 1$ the set $\{1, \ldots, n\}$ can be partitioned into parts of at most three elements whose sums are powers of 3. The partition is given by an explicit recursion.

Keywords: set partition, powers of three, zero-sum triples, involution, formal verification
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1. Introduction

Let $m \geq 2$ and $n \geq 1$ be integers and $[n] = \{1, \ldots, n\}$. Gurvich and Naumova [1] call a partition of $[n]$ *m-good* if every part has at most m elements and the sum of the elements of every part is a power m^k, $k \geq 0$, of m. Every $[n]$ has exactly one 2-good partition [1, Proposition 1]. In an m-good partition every part other than $\{1\}$ has sum $\equiv 0 \pmod{m}$, so $n(n+1)/2 \equiv 0$ or $1 \pmod{m}$. For $n = 2(m^t + 1)$ this sum is $\equiv 3 \pmod{m}$; hence for $m \geq 4$ the set $[n]$ has no m-good partition [1, Observation 1]. For $m = 3$ they stated:

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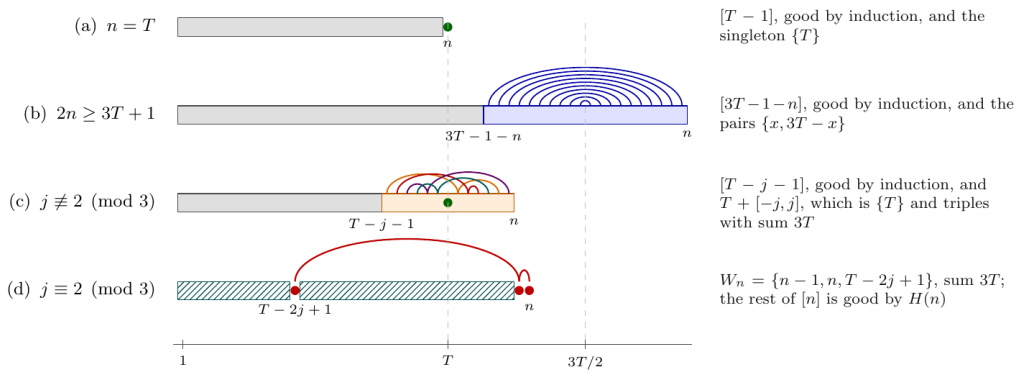


Figure 1: The four cases of the recursion for $T = 27$, drawn for $n = 27, 50, 33, 35$. In (a)–(c) the gray interval is good by induction. In (d) the three red points form W_n, and the rest of $[n]$ is good by $H(n)$, Proposition 4.1.

Conjecture 1.1 ([1, Conjecture 1])**.** *For every $n \geq 1$ the set $[n]$ has a 3-good partition.*

They proved that a minimal counterexample is of the form $n = 3^t + 3k + 2$ with $0 \leq k < (3^{t-1} - 1)/2$ and $k \neq (3^\ell - 1)/2$ for $0 \leq \ell < t$ [1, Theorem 1], recorded further reductions found by O. Polubasoff, and reported that a computation by D. Rybin confirmed the conjecture for $n \leq 844$. We prove it.

Theorem 1.2. *For every $n \geq 1$ the set $[n]$ has a 3-good partition.*

Corollary 1.3. *Let $m \geq 2$. Every set $[n]$, $n \geq 1$, has an m-good partition if and only if $m \in \{2, 3\}$.*

The proof is a recursion on n. Let $T = 3^t \leq n < 3T$. For $n = T$, for $2n > 3T$, and for $n = T + j$ with $j \not\equiv 2 \pmod 3$, the reductions of [1, Section 2] pass from $[n]$ to a shorter interval. For the remaining $n = T + j$, $j \equiv 2 \pmod 3$, $2j < T$, we prove more: $[n]$ minus the triple $\{n-1, n, T-2j+1\}$, whose sum is $3T$, has a 3-good partition (Proposition 4.1). This statement passes from j to n by a reflection argument (Lemma 3.1), except when j lies in the upper half of the interval $[3^u, 3^{u+1})$ that contains it. In that case a fixed-point-free involution of $\{1, \ldots, 2N-2\}$ with distinct pair sums supplies the missing triples (Lemma 3.2). Figure 1 shows the four cases, and Section 5 describes the formal verification of the proof.

2. Good and balanced sets

For integers $a \le b$ let $[a,b] = \{x \in \mathbb{Z} : a \le x \le b\}$, and $[a,b] = \emptyset$ if $a > b$. For $Z \subset \mathbb{Z}$ and $c, s \in \mathbb{Z}$ write $c + sZ = \{c + sz : z \in Z\}$ and $c - Z = \{c - z : z \in Z\}$. A finite set $A \subset \mathbb{Z}$ is *admissible* if $|A| \le 3$ and $\sum_{x \in A} x = 3^k$ for some $k \ge 0$. A finite set $S \subset \mathbb{Z}$ is *good* if it is a disjoint union of admissible sets; thus $[n]$ is good if and only if it has a 3-good partition. A finite set $Z \subset \mathbb{Z}$ is *balanced* if it is a disjoint union of sets each of which is $\{0\}$ or a 3-element set with sum 0. Disjoint unions of good sets are good, and disjoint unions of balanced sets are balanced.

Lemma 2.1 ([1, Lemma 1])**.** *If $\ell \ge 0$ and $\ell \not\equiv 2 \pmod 3$, then $[-\ell, \ell]$ is balanced.*

Proof. Write $\ell = 3k + r$ with $r \in \{0, 1\}$. For $0 \le i < k$ the sets

$$A_i = \{-(3k-i),\ 1+i,\ 3k-1-2i\}, \qquad B_i = \{-(2k-i),\ -(k-i),\ 3k-2i\}$$

have sum 0. As i runs over $[0, k-1]$, the entries $-(3k-i)$, $-(2k-i)$, $-(k-i)$ run over $[-3k, -2k-1]$, $[-2k, -k-1]$, $[-k, -1]$, the entries $1+i$ run over $[1, k]$, and the entries $3k-1-2i$ and $3k-2i$ run over the numbers in $[k+1, 3k]$ of the parity of $3k-1$ and of $3k$. So the $2k$ sets A_i, B_i partition $[-3k, 3k] \setminus \{0\}$. Add $\{0\}$ if $r = 0$ and $\{-\ell, 0, \ell\}$ if $r = 1$. □

In [1] the index i runs from 1; it has to start at 0, as the examples $k = 1, 2$ given there show.

Lemma 2.2. *Let Z be balanced, $q \ge 0$ and $s \ne 0$. Then $3^q + sZ$ is good and sZ is balanced.*

Proof. The map $x \mapsto 3^q + sx$ is injective; it sends $\{0\}$ to $\{3^q\}$ and a 3-element set with sum 0 to a 3-element set with sum 3^{q+1}. The map $x \mapsto sx$ is injective and preserves the two kinds of parts. □

Lemma 2.3. *Let $Q = 3^q$ and let $X \subseteq [1, Q-1]$ satisfy $Q - X = X$. Then X is good.*

Proof. Since Q is odd, $x \ne Q - x$, and X is the disjoint union of the pairs $\{x, Q-x\}$ with $x \in X$, $x < Q/2$; each has sum Q. □

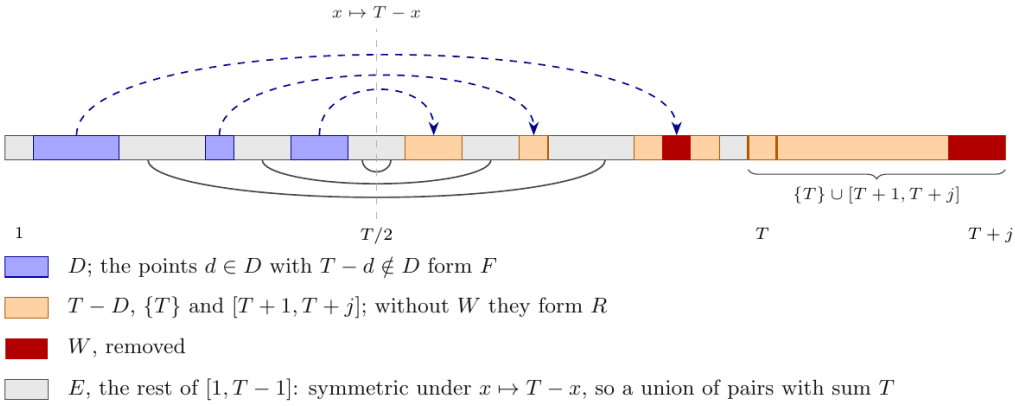


Figure 2: Lemma 3.1. The reflection $x \mapsto T-x$ maps D onto $T-D$; the rest E of $[1, T-1]$ is symmetric and splits into pairs with sum T.

3. Reflection and the involutions σ_N

Lemma 3.1. *Let $T = 3^t$, $j \geq 0$, $D \subseteq [1, T-1]$ and $W \subseteq [T+1, T+j] \cup (T-D)$. Put*

$$F = \{d \in D : T - d \notin D\}, \qquad R = \big(\{T\} \cup [T+1, T+j] \cup (T-D)\big) \setminus W.$$

If F and R are good, then $[1, T+j] \setminus W$ is good.

Proof. Let $E = \{x \in [1, T-1] : x \notin D,\ T-x \notin D\}$. Then $T-E = E$, so E is good by Lemma 2.3. We show that $[1, T+j] \setminus W$ is the disjoint union of E, F and R. The set E meets neither $D \supseteq F$ nor $\{T\} \cup [T+1, T+j] \cup (T-D) \supseteq R$; and $F \subseteq [1, T-1]$ does not meet $T-D$, hence meets neither R nor W. Since E does not meet W either, E, F, R are disjoint subsets of $[1, T+j] \setminus W$. Conversely let $x \in [1, T+j] \setminus W$. If $x \geq T$, or if $x \leq T-1$ and $T - x \in D$, then $x \in R$. Otherwise $x \leq T-1$, $T-x \notin D$, and $x \in F$ or $x \in E$ according as $x \in D$ or not. □

Lemma 3.2. *Let $N \geq 1$, $N \equiv 1 \pmod 3$, and $K = 2N - 2$. There is a fixed-point-free involution σ of $[1, K]$ with the following properties, where $\Pi = \{z \in [1, K] : z < \sigma(z)\}$.*

(i) The sums $z + \sigma(z)$, $z \in \Pi$, are distinct.

(ii) The N numbers $2N - 2$ and $z + \sigma(z) - (2N+1)$, $z \in \Pi$, are distinct, and the set Δ_N they form is balanced.

(iii) Every $\delta \in \Delta_N$ satisfies $|\delta| \le 2N-2$, and every positive $\delta \in \Delta_N$ is even.

Proof. Since $N \equiv 1 \pmod 3$, either $N = 1$, or $N = 2M$ with $M \ge 2$ and $M \equiv 2 \pmod 3$, or $N = 2M+1$ with $M \ge 3$ and $M \equiv 0 \pmod 3$.

If $N = 1$, then $[1, K] = \emptyset$ and $\Delta_1 = \{0\}$.

Let $N = 2M$. Let σ exchange $z \leftrightarrow z + N - 1$ for $1 \le z \le N-1$. These pairs partition $[1, K]$, since $K = 2(N-1)$. The pair sums are $2z + N - 1$, $1 \le z \le N-1$, which are distinct. Subtracting $2N + 1 = 4M + 1$ gives the numbers $2(z - M - 1)$, $1 \le z \le 2M-1$, so $\Delta_N = \{4M-2\} \cup 2\,[-M, M-2]$, that is,

$$\Delta_N = \{4M-2,\ -2M,\ -(2M-2)\} \cup 2\,[-(M-2), M-2].$$

The first set has sum 0; the second lies in $[-(2M-4), 2M-4]$, so it does not meet the first, and it is balanced by Lemmas 2.1 and 2.2, since $M - 2 \equiv 0 \pmod 3$. This proves (i)–(iii), as $4M - 2 = 2N - 2$.

Let $N = 2M+1$. Let σ exchange $1 \leftrightarrow N$, $2 \leftrightarrow N+1$, $3 \leftrightarrow N-1$, $4 \leftrightarrow N+2$, $2N-3 \leftrightarrow 2N-2$, and $x \leftrightarrow x+N-2$ for $5 \le x \le N-2$. The last family pairs $[5, N-2]$ with $[N+3, 2N-4]$, so these pairs partition $[1, K]$. The pair sums are $N+1$, $N+3$, $N+2$, $N+6$, $4N-5$ and $2x+N-2 \in [N+8, 3N-6]$, $5 \le x \le N-2$, which are distinct. Subtracting $2N + 1 = 4M + 3$ gives

$$\Delta_N = \{4M,\ -(2M+1),\ -(2M-1)\} \cup \{4M-4,\ -2M,\ -(2M-4)\} \cup 2\,[-(M-3), M-3].$$

The first two sets have sum 0; the third lies in $[-(2M-6), 2M-6]$ and is balanced since $M - 3 \equiv 0 \pmod 3$. The three sets are disjoint, which proves (i)–(iii), as $4M = 2N - 2$. □

4. Proof of Theorem 1.2

For $n \ge 1$ let t be the integer with $3^t \le n < 3^{t+1}$. If $t \ge 1$ and $j = n - 3^t$ satisfies

$$j \equiv 2 \pmod 3, \qquad 2 \le j, \qquad 2j + 1 \le 3^t, \tag{1}$$

put $W_n = \{n-1,\ n,\ 3^t - 2j + 1\}$. Its elements are distinct, since $2 \le 3^t - 2j + 1 \le 3^t - 3$, and their sum is 3^{t+1}. Let $H(n)$ be the statement that $[n] \setminus W_n$ is good.

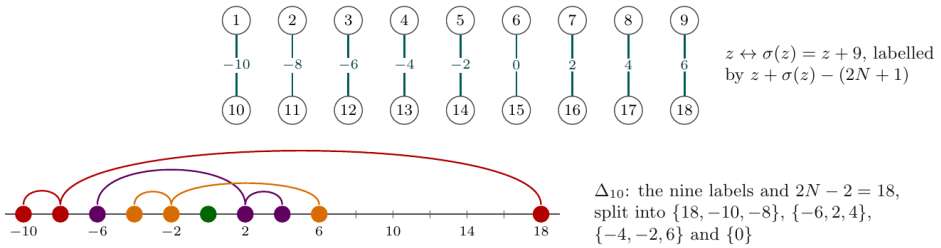


Figure 3: Lemma 3.2 for $N = 10$, $M = 5$. Each pair $\{z, \sigma(z)\}$ carries the number $z + \sigma(z) - (2N + 1)$; together with $2N - 2 = 18$ these form Δ_{10}, which splits into three triples with sum 0 and $\{0\}$.

Proposition 4.1. *Let $n = 3^t + j$ with $t \geq 1$ and j as in (1). Suppose that $[m]$ is good for every $m < n$ and that $H(m)$ holds for every $m < n$ for which it is defined. Then $H(n)$ holds.*

Proof. Let $T = 3^t$, so $W_n = \{T+j-1, T+j, T-2j+1\}$. Since $T \geq 2j+1 \geq 5$, we have $T \geq 9$. In each case below we choose $D \subseteq [1, T-1]$ with $2j-1 \in D$, so that $W = W_n \subseteq [T+1, T+j] \cup (T - D)$, and show that the sets F and R of Lemma 3.1 are good. Lemma 3.1 then gives $H(n)$.

Case $j = 2$. Take $D = \{3\}$. Then $F = \{3\}$, as $T - 3 \neq 3$, and $R = \{T, T+1, T+2, T-3\} \setminus W = \{T\}$.

For $j \geq 5$ let u be the integer with $3^u \leq j < 3^{u+1}$. Then $u \geq 1$, and $j \geq 3^u + 2$ because $j \equiv 2$ and $3^u \equiv 0 \pmod 3$. Also $3j \neq T$, since $3j = T$ would give $j = 3^{t-1} \equiv 0 \pmod 3$.

Cases $3j > T$, and $3j < T$ with $2j + 1 \leq 3^{u+1}$. Take $D = [1, j-2] \cup \{2j-1\}$. Then $T - D = [T-j+2, T-1] \cup \{T-2j+1\}$ and

$$R = T + [-(j-2), j-2],$$

which is good by Lemmas 2.1 and 2.2, since $j - 2 \equiv 0 \pmod 3$.

If $3j > T$, then $u = t - 1$. As T and $3j$ are multiples of 3, $T \leq 3j - 3$, so $T-2j+1 \in [2, j-2] \subseteq D$. For $d \in [1, j-2]$ we have $T-d \geq T-j+2 > j-2$, so $T - d \in D$ only if $T - d = 2j - 1$. Hence

$$F = [1, j-2] \setminus \{T - 2j + 1\} = [j] \setminus W_j,$$

because $3^{u+1} = T$ and $W_j = \{j-1, j, T-2j+1\}$. The number $j = 3^u + (j - 3^u)$ satisfies (1) at level u, so F is good by $H(j)$.

If $3j < T$ and $2j+1 \le 3^{u+1}$, then $3^{u+1} \le 3j < T$ gives $T \ge 3^{u+2}$, and every $d \in D$ satisfies $T - d \ge T - 2j + 1 > 3^{u+2} - 3^{u+1} > 2j - 1$. Hence $F = D$. Put $p = 3^{u+1} - 2j + 1$; then $2 \le p \le j - 2$, because $2j + 1 \le 3^{u+1}$ and $j \ge 3^u + 1$. Since $W_j = \{j-1, j, p\}$,

$$F = D = \big([j] \setminus W_j\big) \cup \{p,\, 2j-1\},$$

a disjoint union of a set that is good by $H(j)$ and a pair with sum 3^{u+1}.

Case $3j < T$ *and* $2j + 1 > 3^{u+1}$. Let $P = 3^{u+1}$. As before $T \ge 3P$, so $T = 3P$ or $T \ge 9P$; and $P + 1 \le 2j \le 2P - 2$. Write $P = 2c + 1$ and put

$$N = j - c, \qquad e = P - 1 - j = c - N.$$

Then $N \ge 1$, $e \ge 0$, $j = e + 2N$, and $2N - 2 = 2j - P - 1 \le P - 3$. Since $3 \mid P$ we have $c \equiv 1 \pmod 3$, hence $N \equiv 1$ and $e \equiv 0 \pmod 3$. Let σ, Π, Δ_N be as in Lemma 3.2, $K = 2N - 2$, and

$$D = [1, e] \cup (P + \Delta_N).$$

By Lemma 3.2(iii), $P + \Delta_N \subseteq [P - 2N + 2, P + 2N - 2] = [2e + 3, 2j - 1]$, so the union is disjoint, $D \subseteq [1, 2P - 3] \subseteq [1, T - 1]$, and $2j - 1 = P + (2N - 2) \in D$.

We show $F = D$. Let $d, d' \in D$ with $d + d' = T$. If $d \le e$, then $d + d' \le e + 2P - 3 < 3P \le T$; so $d = P + \delta$ and $d' = P + \delta'$ with $\delta, \delta' \in \Delta_N$ and $\delta + \delta' = T - 2P$. If $T \ge 9P$, then $\delta + \delta' \ge 7P > 2(P - 3)$, which is impossible. If $T = 3P$, then $\delta + \delta' = P$ is odd. Since $\delta, \delta' \le P - 3$, both are positive, hence even by Lemma 3.2(iii), a contradiction. Thus $F = D$, which is good: $[1, e] = [e]$ is good by hypothesis, and $P + \Delta_N$ by Lemmas 3.2 and 2.2.

It remains to compute R. First, $[T + 1, T + j] \setminus W = T + [1, j - 2] = T + [1, e + K]$. Second, $T - D = T + [-e, -1]$ together with $T - P - \Delta_N$; here $T - P - (2N - 2) = T - 2j + 1 \in W$, and for $z \in \Pi$, using $P - 2N - 1 = 2e$,

$$T - P - \big(z + \sigma(z) - (2N + 1)\big) = T - \big(2e + z + \sigma(z)\big).$$

By Lemma 3.2(ii) these numbers differ from $T - 2j + 1$. Hence $R = T + Z$ with

$$Z = [-e, e] \;\cup \bigcup_{z \in \Pi} \big\{e + z,\; e + \sigma(z),\; -(2e + z + \sigma(z))\big\}.$$

Each set $\{e + z, e + \sigma(z), -(2e + z + \sigma(z))\}$ has three elements and sum 0. As z runs over Π, the numbers $e + z$ and $e + \sigma(z)$ run over $e + [1, K]$ once

each, because σ is a fixed-point-free involution; the numbers $-(2e+z+\sigma(z))$ are distinct by Lemma 3.2(i) and less than $-e$. Since $[-e,e]$ is balanced by Lemma 2.1, Z is balanced, and R is good by Lemma 2.2. □

Proof of Theorem 1.2. We prove by strong induction on $n \geq 0$ that $[n]$ is good and that $H(n)$ holds whenever it is defined; $[0] = \emptyset$. The sets $[1] = \{1\}$ and $[2] = \{1,2\}$ are admissible, and $H(1)$, $H(2)$ are not defined. Let $n \geq 3$, $T = 3^t \leq n < 3T$, $t \geq 1$. Whenever $H(n)$ is defined, it holds by Proposition 4.1 and the induction hypothesis. For $[n]$ there are four cases.

(a) $n = T$: $[n] = [T-1] \cup \{T\}$.
(b) $2n \geq 3T+1$: with $a = 3T-1-n < n$, $[n] = [a] \cup [a+1, 3T-1-a]$, and the second set is good by Lemma 2.3 with $Q = 3T$.
(c) $T < n$, $2n \leq 3T$ and $j = n - T \not\equiv 2 \pmod 3$: $[n]$ is the union of $[T-j-1]$ and $T + [-j,j]$, and the second set is good by Lemmas 2.1 and 2.2.
(d) $T < n$, $2n \leq 3T$ and $j = n - T \equiv 2 \pmod 3$: then j satisfies (1), and $[n] = ([n] \setminus W_n) \cup W_n$, where $[n] \setminus W_n$ is good by $H(n)$ and W_n is admissible.

In (a)–(c) the first set is good by the induction hypothesis, and in all cases the union is disjoint. □

Proof of Corollary 1.3. For $m = 2$ this is [1, Proposition 1], for $m = 3$ it is Theorem 1.2, and for $m \geq 4$ the set $[2m+2]$ has no m-good partition, as recalled in Section 1. □

Example 4.2. The least n not excluded by [1, Theorem 1] is $n = 35 = 27+8$. Here $3j = 24 < 27$, $u = 1$ and $2j+1 = 17 > 9$, so the last case applies with $P = 9$, $c = 4$, $N = 4$, $e = 0$. The involution σ of $[1,6]$ is $1 \leftrightarrow 4$, $2 \leftrightarrow 5$, $3 \leftrightarrow 6$, $\Delta_4 = \{6, -4, -2, 0\}$ and $D = \{5, 7, 9, 15\}$. The construction gives the 3-good partition of $[35]$ into

$$\{5,7,15\},\ \{9\},\ \{27\},\ \{18,30,33\},\ \{20,29,32\},\ \{22,28,31\},\ \{12,34,35\}$$

and the nine pairs $\{x, 27-x\}$, $x \in \{1,2,3,4,6,8,10,11,13\}$; see Figure 4.

The recursion computes the partition of $[n]$ in $O(n)$ arithmetic operations.

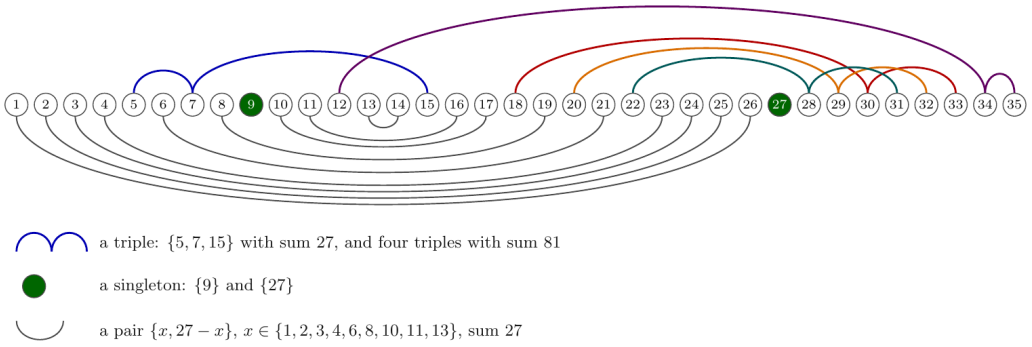


Figure 4: The 3-good partition of [35] given by the recursion. Each triple is drawn as two arcs above the line, each pair as one arc below it.

5. Formal verification

Theorem 1.2 is formalized in Lean 4 [2] with Mathlib [3], version 4.34.1, and the formal proof is checked by the Lean kernel. The formal statement says that for every natural number n there is a partition of $\{1, \dots, n\}$ into nonempty parts, each of which has at most three elements and a power of 3 as its sum. In the formal proof, good and balanced sets are the finite sets of integers built by disjoint union from the admissible sets, respectively from $\{0\}$ and the 3-element sets with sum 0, and the argument is that of Sections 2–4. The proof depends only on the axioms of Lean's logic, namely propositional extensionality, choice and the soundness of quotients; it has no unproved step and adds no axiom.

Independent computations in exact integer arithmetic run the recursion of Section 4 and confirm that it gives a 3-good partition of $[n]$ for every $n \le 30000$ and for further values of n up to $6.5 \cdot 10^7$, and that $H(n)$ holds for every n in these ranges for which it is defined. They also confirm Lemma 2.1 for $\ell \le 30000$ and Lemma 3.2 for $N \le 30001$.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with the proofs, the Lean formalization, the computations and the preparation of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

Deep Bhattacharjee computed the Lean formalization and the Julia, C, C++, shell and Python computations with the help of AI assistance (Claude, Anthropic). The Lean formalization of the proofs in this paper is archived at Zenodo, doi:10.5281/zenodo.0000000. The paper is available at `https://github.com/creelie/didactic-lamp/blob/main/preprintGurvichNaumovaConjecture/GurvichNaumovaConjecture.pdf`.

Competing interests

None.

Funding

None.

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